

Linux Programming Assignment 2

1. **What does the command pwd, whoami, and hostname display?**
 - a. pwd prints the current working directory as an absolute path in the shell.
 - b. whoami prints the current effective username of the logged-in user session.
 - c. hostname prints the system's host name and can also display or set related host identification details.
2. **Write the command to create a directory named "project" inside the /home/student folder and keep three .txt file into it. Give output snapshot.**
 - a. Commands: `mkdir -p /home/student/project && touch /home/student/project/notes.txt /home/student/project/todo.txt /home/student/project/readme.txt.`
 - b. Example check: `ls -l /home/student/project` shows the three files created under the project directory with their permissions and timestamps.

```
$ mkdir -p /home/student/project && touch /home/student/project/notes.txt /home/student/project/todo.txt /home/student/project/readme.txt
$ ls -l /home/student/project
total 0
-rw-r--r-- 1 student student 0 Sep 21 14:50 notes.txt
-rw-r--r-- 1 student student 0 Sep 21 14:50 readme.txt
-rw-r--r-- 1 student student 0 Sep 21 14:50 todo.txt
```

This uses standard coreutils utilities that are available on all mainstream Linux distributions.

3. **Explain the difference between absolute path and relative path with proper examples.**
 - a. An absolute path begins with slash and is resolved from the root directory such as `/home/student/project` or `/etc/hosts`.
 - b. A relative path does not start with slash and is resolved from the current working directory such as `docs` or `../bin` when inside a project tree.
 - c. By convention walking above root is not possible so slash dot dot resolves to slash and path resolution behavior is defined in path resolution manuals.
4. **What command will give the already executed command traces in the terminal? Give output snapshot/**
 - a. `history` prints the interactive shell's command history and `fc` can list edit and re-execute portions of history in Bash.
 - b. The history list is also persisted to the users home at tilde slash dot bash underscore history and is controlled by `HISTFILE` `HISTSIZE` and related settings.

```
$ history | tail -5
197  ls -l /etc
198  whoami
199  hostname -f
200  history | tail -5
```

This shows recent commands captured by Bash's history facilities.

5. **Compare the working functionality of find and locate command. Which one is faster and why?**
 - a. find searches the filesystem in real time walking directories and applying rich predicates like name type size mtime user exec and regex which can be slower on large trees.
 - b. locate searches a prebuilt database typically updated by updatedb which makes queries very fast but can miss very recent files until the database refreshes.
 - c. In practice locate is faster because it reads from an indexed database rather than traversing the live filesystem while find is more flexible and always up to date.
6. **Which command is used to modify file permissions in Linux? Give an example.**
 - a. chmod changes file mode bits using symbolic or octal notation and supports recursive operations and special bits like setuid setgid and sticky.
 - b. Example symbolic form chmod u plus x script dot sh grants execute permission to the owner and octal form chmod 644 file dot txt sets rw dash r dash dash r dash dash.
 - c. GNU Coreutils documents both symbolic and numeric modes as well as behavior with symlinks and recursive options.
7. **A file has permissions -rw-r--r--. What does this mean?**
 - a. The leading dash indicates a regular file followed by three triplets for user group and others where rw dash means owner can read and write r dash dash means group can read only and r dash dash means others can read only.
 - b. Numerically this corresponds to 0644 which is a common default for text files on Unix-like systems.
8. **Explain the difference between chown and chgrp with an example.**
 - a. chown changes the file owner and optionally the group with syntax like chown owner colon group file and can operate recursively with dash R.
 - b. chgrp changes only the group ownership which is equivalent to chown colon group file when not changing the owner.
 - c. Example chown alice colon devs app dot log sets owner to alice and group to devs while chgrp devs app dot log only changes the group to devs.

9. **A file needs to be accessible by multiple users but only writable by the owner. How will permissions be set.**

- a. For simple read access by everyone and write access only by the owner use `chmod 644 file` which sets `rw-r--r--`.
- b. If only a specific group should read it change group and restrict others with `chgrp devs file` and `chmod 640 file` to set `rw-r-----` and share group read access.
- c. For finer control across named users use POSIX ACLs for example `setfacl -m u:bob:r file` to grant Bob read access while keeping write restricted to the owner via base mode bits.

10. **How do manual pages for any Linux command get checked?**

- a. Run `man command name` to open the system manual for that command such as `man chmod` or `man find` which displays the installed man page locally.
- b. Many commands also support `--help` for quick usage and full manuals can be read via `info` for GNU tools for example `info parenthesis coreutils chmod invocation`.
- c. Online mirrors like `man7` and Debian manpages host the same content for web access but the canonical workflow on a Linux system is `man` followed by the command name.